
Supplementary information

**Inter-brain neural dynamics in biological
and artificial intelligence systems**

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Supplementary Information for

Inter-brain neural dynamics in biological and artificial intelligence systems

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Supplementary Table 1. Detailed statistical information.

Figure	Sample size	Statistical test	Test statistic and P value
Fig. 1j	N = 13 pairs of male mice	Two-tailed Wilcoxon matched-pairs signed rank test	W = -91.0, ***P = 0.0002
Fig. 1m	N = 14 pairs of male mice	Two-tailed Wilcoxon matched-pairs signed rank test	W = -105.0, ***P = 0.0001
Fig. 1l	N = 13 pairs of male mice in glutamatergic	Two-way repeated measures analysis of variance (ANOVA) with post hoc Sidák's multiple comparison test	ANOVA: Signal Type (Glutamatergic vs Phase Randomized): F(1, 120) = 46.04, ***P < 0.0012;
			Frequency band: F(4, 120) = 5.203, ***P = 0.0007;
			Interaction: F(4,120) = 5.087, ***P = 0.0008
			<1: t(120) = 1.497, P = 0.5213 1-15: t(168) = 1.119, P = 0.7860 15-30: t(168) = 3.035, *P = 0.0147 30-60: t(168) = 2.717, *P = 0.0373 >60: t(168) = 6.805, ***P < 0.0001
Fig. 1o	N = 14 pairs of male mice in GABAergic	Two-way repeated measures analysis of variance (ANOVA) with post hoc Sidák's multiple comparison test	ANOVA: Signal Type (GABAergic vs Phase Randomized): F(1, 130) = 243.6, ***P < 0.0001;
			Frequency band: F(4, 130) = 11.44, ****P < 0.0001;
			Interaction: F(4, 130) = 11.7, ****P < 0.0001
			<1: t(130) = 2.293, P = 0.1118 1-15: t(130) = 5.019, ****P < 0.0001 15-30: t(130) = 7.23, ****P < 0.0001 30-60: t(130) = 9.692, ****P < 0.0001 >60: t(130) = 10.67, ****P < 0.0001
Fig. 1q	N = 9 mice in GABAergic, N = 8 mice in glutamatergic	Mann-Whitney unpaired two-tailed test	Attack-Sniff: R1 = 117, R2 = 36; U = 0, ****P < 0.0001
Fig. 1r	N = 5 mice in GABAergic, N = 8 mice in glutamatergic	Mann-Whitney unpaired two-tailed test	Attack-Approach: R1 = 55, R2 = 36; U = 0, **P = 0.0016
Fig. 1s	N = 8 mice in GABAergic, N = 5 mice in glutamatergic	Mann-Whitney unpaired two-tailed test	Attack-Self-groom: R1 = 71, R2 = 20; U = 5, *P = 0.0295
Fig. 1t	N = 27 mice in GABAergic, N = 23 mice in glutamatergic	Mann-Whitney unpaired two-tailed test	Sniffing-Self-groom: R1 = 899, R2 = 376; U = 100, ****P < 0.0001
Fig. 1u	N = 14 pairs in GABAergic	Friedman test and Dunn's multiple comparison test	Friedman statistic = 36.60, ****P < 0.0001
			Original-Social Z = 3.367, **P = 0.0023
			Original-Shuffled: Z = 1.464, P = 0.4297
			Original-Non-social: Z = 2.489, *P = 0.0385
Fig. 1v	N = 13 pairs in glutamatergic	Friedman test and Dunn's multiple comparison test	Friedman statistic = 20.82, ****P = 0.0001
			Original-Social: Z = 1.215, P = 0.6728
			Original-Shuffled: Z = 1.215, P = 0.6728
			Original-Non-social: Z = 2.734, *P = 0.0187
Fig. 2g	N = 14 pairs of male mice in GABAergic	Two-tailed Wilcoxon matched-pairs signed rank test	GABAergic Interaction vs Separation: W = -105.0, ***P = 0.0001
Fig. 2h	N = 13 pairs of male mice in glutamatergic	Two-tailed Wilcoxon matched-pairs signed rank test	Glutamatergic Interaction vs Separation: W = -66.0, ***P = 0.001
Fig. 2i	N = 14 pairs of male mice in GABAergic, N = 13 pairs of male mice in glutamatergic	Two-tailed Mann Whitney test	GABAergic-glutamatergic: R1 = 285, R2 = 93, U = 2, ****P < 0.0001
Fig. 2j		Two-tailed Mann Whitney test	GABAergic-glutamatergic: R1 = 212, R2 = 166, U = 75, P = 0.4583
Fig. 2m			ANOVA:

	N = 28 male mice in GABAergic social and Non-social N = 26 male mice in glutamatergic social and Non-social	Two-way ANOVA with uncorrected Fisher's LSD	Cell type: F(1, 104) = 216.1, ****P < 0.0001 Interaction session-Separation session: F(1, 104) = 28.87, ****P < 0.0001 Interaction: F(1, 104) = 31.51, ****P < 0.0001 GABAergic social-glutamatergic social: t = 14.36, ****P < 0.0001 GABAergic Non-social-glutamatergic Non-social: t = 6.424, ****P < 0.0001 GABAergic social-Non-social: t = 0.1735, P = 0.8626 Glutamatergic social-Non-social: t = 7.628, ****P < 0.0001
Fig. 2o	N = 28 male mice in GABAergic N = 26 male mice in glutamatergic	Two-tailed Mann Whitney test	GABAergic-glutamatergic PLSC1: R1 = 1020, R2 = 255, U = 2, ****P < 0.0001
Fig. 2p	N = 28 male mice in GABAergic	Two-tailed Wilcoxon matched-pairs signed rank test	W = -340.0, ****P < 0.0001
Fig. 2q	across all groups		W = 292.0, ***P = 0.0005
Fig. 2r	N = 14 pairs of male mice across all groups	Two-tailed Mann Whitney test	Low mutual social vs high mutual social: R1 = 261, R2 = 145, U = 40, **P = 0.0067
Fig. 2s	N = 28 mice in GABAergic neurons		
Fig. 2t	N = 14 pairs of male mice in GABAergic	Linear regression	F = 25.58, ***P = 0.0003
Fig. 3a	N = 14 pairs of male mice across all groups	Friedman test and Dunn's multiple comparisons test	Q = 77.91, ****P < 0.0001 Social-Social shuffled: Z = 8.074, ****P < 0.0001 Non-social-Non-social shuffled: Z = 3.519, **P = 0.0026
Fig. 3b	N = 28 mice across all groups	Friedman test and Dunn's multiple comparisons test	Q = 80.61, ****P < 0.0001 Social-Social shuffled: Z = 8.384, ****P < 0.0001 Non-social-Non-social shuffled: Z = 3.209, **P = 0.0080
Fig. 3c	N = 14 pairs of male mice	Linear regression	Interaction: F = 5.516, *P = 0.0368
Fig. 3d		Linear regression	Separation: F = 2.850, P = 0.1172
Fig. 3e		Linear regression	Social moments: F = 15.39, **P = 0.0020
Fig. 3f		Linear regression	Non-social moments: F = 0.04930, P = 0.8288
Fig. 3j	Interaction session: N = 8950 ensemble pairs in 28 male mice	Two-tailed Wilcoxon matched-pairs signed rank test	W = -30026405, ****P < 0.0001
Fig. 3k	Separation session: N = 7579 ensemble pairs in 28 male mice	Two-tailed Wilcoxon matched-pairs signed rank test	W = 14196186, ****P < 0.0001
Fig. 3l	Social moments: N = 6246 ensemble pairs in 28 male mice	Two-tailed Wilcoxon matched-pairs signed rank test	W = -12827925, ****P < 0.0001
Fig. 3m	Non-social moments: N = 9112 ensemble pairs in 28 pairs of male mice	Two-tailed Wilcoxon matched-pairs signed rank test	W = 12459450, ****P < 0.0001
Fig. 3p	N = 14 pairs of male mice across all groups	Friedman test and Dunn's multiple comparisons test	Q = 23.29, ****P < 0.0001 Original-x-Interaction: Z = 3.213, **P = 0.0026 Original-x-Temporal Coupling: Z = 4.725, ****P < 0.0001
Fig. 4c	N = 54 across all groups	Two-tailed Wilcoxon matched-pairs signed rank test	W = -1485, ****P < 0.0001
Fig. 4e	N = 27 across all groups	Friedman test and Dunn's multiple comparisons test	Q = 48.08, ****P < 0.0001 Observed-Shuffled: Z = 4.899, ****P < 0.0001 Observed-Cross-Pair: Z = 5.103, ****P < 0.0001
Fig. 4i	N = 28 across all groups	Friedman test and Dunn's multiple comparisons test	Q = 41.14, ****P < 0.0001 Shared vs Unique: Z = 6.414, ****P < 0.0001 Shared vs Full: Z = 3.207, **P = 0.0040 Unique vs Full: Z = 3.207, **P = 0.0040
Fig. 4k	N = 28 across all groups		
Fig. 4l			
Fig. 4n			
Fig. 4o			No statistical comparisons
Fig. 5i	N = 10 for Social agents N = 10 for Non-social agents	Two-tailed Mann Whitney test	Non-social-Social: R1=155, R2=55, U=0, ****P < 0.0001
Fig. 5j		Two-tailed Mann Whitney test	Non-social-Social: R1=155, R2=55, U=0, ****P < 0.0001
Fig. 5k		Two-tailed Mann Whitney test	Non-social-Social: R1=155, R2=55, U=0, ****P < 0.0001
Fig. 5l		Two-tailed Mann Whitney test	Non-social-Social: R1=155, R2=55, U=0, ****P < 0.0001
Fig. 5m		Two-tailed Mann Whitney test	Non-social-Social: R1=155, R2=55, U=0, ****P < 0.0001
Fig. 5n		Two-tailed Mann Whitney test	Non-social-Social: R1=155, R2=55, U=0, ****P < 0.0001
Fig. 5r			ANOVA: Angle: F(5,54) = 228.0, ****P < 0.0001

			Training stage: $F(1,54) = 1.24e-7$, $P = 0.9997$ Interaction: $F(5,54) = 144.0$, **** $P < 0.0001$ 1-30: $t = 16.47$, **** $P < 0.0001$ 31-60: $t = 13.62$, **** $P < 0.0001$ 61-90: $t = 2.847$, * $P = 0.0368$ 91-120: $t = 6.864$, **** $P < 0.0001$ 121-150: $t = 10.66$, **** $P < 0.0001$ 151-180: $t = 9.713$, **** $P < 0.0001$
Fig. 6b	N = 10 across all groups	Two-tailed Mann Whitney test	Non-social-Social: $R1=155$, $R2=55$, $U=0$, **** $P < 0.0001$
Fig. 6c		Two-tailed Mann Whitney test	Non-social-Social: $R1=154$, $R2=56$, $U=1$, **** $P < 0.0001$
Fig. 6d		Two-tailed Mann Whitney test	Non-social-Social: $R1=155$, $R2=55$, $U=0$, **** $P < 0.0001$
Fig. 6e		Two-tailed Mann Whitney test	Non-social-Social: $R1=148$, $R2=62$, $U=7$, *** $P = 0.0005$
Fig. 6g	N = 56 pairs in Social, N =59 pairs in Non-social, N =59 pairs in Non-social with vision	Kruskal-Wallis test and Dunn's multiple comparison test	Kruskal-Wallis: $H=153.9$, **** $P < 0.0001$
			Social-Non-social: $Z = 11.68$, **** $P < 0.0001$
			Social-Non-social with vision: $Z = 9.228$, **** $P < 0.0001$
			Non-social-Non-social with vision: $Z = 2.482$, $P = 0.0783$
Fig. 6h		Kruskal-Wallis test and Dunn's multiple comparison test	Kruskal-Wallis: $H=167.2$, **** $P < 0.0001$
			Social-Non-social: $Z = 12.64$, **** $P < 0.0001$
			Social-Non-social with vision: $Z = 8.679$, **** $P < 0.0001$
			Non-social-Non-social with vision: $Z = 4.009$, *** $P = 0.0004$
Fig. 6k	N = 55 across all groups	Friedman test and Dunn's multiple comparisons test	$Q = 106.3$, **** $P < 0.0001$ Original-xIntraset: $Z = 10.30$, **** $P < 0.0001$ Original-xTemporal Coupling: $Z = 5.575$, **** $P < 0.0001$
Fig. 6l	N = 112 in Non-social, N = 118 in Social	Two-tailed Mann Whitney test	Non-social-Social: $R1=19544$, $R2=7021$, $U=0$, **** $P < 0.0001$
Fig. 6m		Two-tailed Mann Whitney test	Non-social-Social: $R1=18989$, $R2=7576$, $U=555$, **** $P < 0.0001$
Fig. 6n	N = 56 pairs of Social agents	Two-tailed Wilcoxon matched-pairs signed rank test	$W = 700.0$, ** $P = 0.0038$
Fig. 6o	N = 14 pairs of male mice	Two-tailed Wilcoxon matched-pairs signed rank test	$W = -89.00$, ** $P = 0.0031$
Fig. 6q	N = 219	Linear regression	$F = 129.0$, **** $P < 0.0001$
Fig. 6r	N = 220	Linear regression	$F = 0.6615$, $P = 0.4169$
Fig. 6s	N = 10 across all groups	Friedman test and Dunn's multiple comparisons test	$Q = 15.8$, **** $P < 0.0001$
			PLSCs-Rand PCs: $Z = 2.907$, * $P = 0.011$
			PLSCs-Original: $Z = 3.801$, **** $P = 0.0004$
			Rand PCs-Original: $Z = 0.8944$, $P > 0.9999$
Fig. 6t		Friedman test and Dunn's multiple comparisons test	$Q = 18.2$, **** $P < 0.0001$
			PLSCs-Rand PCs: $Z = 2.46$, * $P = 0.0417$
			PLSCs-Original: $Z = 4.249$, **** $P < 0.0001$
			Rand PCs-Original: $Z = 1.789$, $P = 0.2209$
Fig. 6u		Friedman test and Dunn's multiple comparisons test	$Q = 18.2$, **** $P < 0.0001$
			PLSCs-Rand PCs: $Z = 2.46$, * $P = 0.0417$
			PLSCs-Original: $Z = 4.249$, **** $P < 0.0001$
			Rand PCs-Original: $Z = 1.789$, $P = 0.2209$
Extended Data Fig. 1e	N = 14 pairs of male mice in GABAergic Interaction and Separation. N = 28 random pairs in Cross-pair control	Kruskal Wallis test followed by Dunn's multiple comparison test	**** $P < 0.0001$, $H=29.98$ Interaction-Separation: **** $P < 0.0001$, $Z = 4.994$ Interaction-Cross-pair: **** $P < 0.0001$, $Z = 4.716$ Separation-Cross-pair: $P = 0.2936$, $Z = 1.050$
Extended Data Fig. 1g	N = 13 pairs of male mice in glutamatergic Interaction and Separation. N = 26 random pairs in Cross-pair control.	Kruskal Wallis test followed by Dunn's multiple comparison test	*** $P = 0.0005$, $H=15.05$ Interaction-Separation: ** $P = 0.0068$, $Z = 2.705$ Interaction-Cross-pair: *** $P = 0.0001$, $Z = 3.833$ Separation-Cross-pair: $P = 0.4778$, $Z = 0.7098$
Extended Data Fig. 1k	N = 13 pairs of male mice in glutamatergic. N = 14 pairs of male mice in GABAergic.	Two-way repeated measures analysis of variance (ANOVA) with post hoc Sidák's multiple comparison test	ANOVA: Signal Type (GABAergic vs Glutamatergic): $F(1, 175) = 43.16$, **** $P < 0.0001$; Frequency band: $F(6, 175) = 10.01$, **** $P < 0.0001$; Interaction: $F(6,175) = 1.309$, $P = 0.2555$ <1: $t(175) = 0.4524$, $P = 0.6516$ 1-15: $t(175) = 2.201$, * $P = 0.0290$ 15-30: $t(175) = 2.343$, * $P = 0.0202$ 30-45: $t(175) = 4.133$, **** $P < 0.0001$ 45-60: $t(175) = 3.318$, ** $P = 0.0011$ 60-120: $t(175) = 2.793$, ** $P = 0.0058$ >120: $t(175) = 2.142$, * $P = 0.0336$

Extended Data Fig. 1l	N = 14 pairs of male mice in GABAergic across all groups.	Two-way repeated measures analysis of variance (ANOVA)	ANOVA: Signal Type (Interaction vs Separation): F(1, 130) = 100.1, ****P < 0.0001
Extended Data Fig. 1m	N = 13 pairs of male mice in glutamatergic across all groups.	Two-way repeated measures analysis of variance (ANOVA)	ANOVA: Signal Type (Interaction vs Separation): F(1, 120) = 7.243, **P = 0.0081
Extended Data Fig. 1o	N = 14 pairs of male mice in GABAergic across all groups.	Friedman test followed by Dunn's multiple comparison test	Q = 18.43, ****P < 0.0001
			Interaction-Separation: Z = 3.402, ***P = 0.0007
			Interaction-Shuffle: Z = 3.969, ****P < 0.0001
			Separation-Shuffle: Z = 0.5669, P = 0.5708
Extended Data Fig. 1o	N = 13 pairs of male mice in glutamatergic across all groups.	Friedman test followed by Dunn's multiple comparison test	Q = 10.31, **P = 0.0058
			Interaction-Separation: Z = 2.157, *P = 0.0310
			Interaction-Shuffle: Z = 3.138, ***P = 0.0017
			Separation-Shuffle: Z = 0.9806, P = 0.3268
Extended Data Fig. 1r	N = 14 pairs of male mice in GABAergic across all groups.	Two-way repeated measures analysis of variance (ANOVA)	ANOVA: Signal Type (Interaction vs Separation): F(1, 130) = 58.38, ****P < 0.0001
Extended Data Fig. 1s	N = 13 pairs of male mice in glutamatergic across all groups.	Two-way repeated measures analysis of variance (ANOVA)	ANOVA: Signal Type (Interaction vs Separation): F(1, 120) = 3.747, P = 0.0553
Extended Data Fig. 1t	N = 3331 GABAergic neurons in 14 pairs of male mice and 9006 glutamatergic neurons in 13 pairs of male mice	Two-tailed Mann-Whitney test	Firing Rates of GABAergic versus glutamatergic neurons: R1=31117463, R2=44989490, U=4430969, ****P < 0.0001
Extended Data Fig. 1u	N = 3331 GABAergic neurons in 14 pairs of male mice and 1798 glutamatergic neurons in 13 pairs of male mice	Two-tailed Mann-Whitney test	Firing Rates of GABAergic versus glutamatergic neurons: R1=8560768, R2=4595117, U=2977816, P = 0.7406
Extended Data Fig. 1v	N = 14 pairs of male mice GABAergic group and 13 pairs of male mice in glutamatergic group	Two-tailed Mann-Whitney test	Inter-brain PCC of GABAergic versus glutamatergic neurons: R1=287, R2=91, U=0, ****P < 0.0001
Extended Data Fig. 1w	N = 1674 GABAergic neurons in 14 pairs of male mice and 4507 glutamatergic neurons in 13 pairs of male mice	Two-tailed Mann-Whitney test	Firing Rates of GABAergic versus glutamatergic neurons: R1=5399043, R2=13706428, U=3547650, ***P = 0.0003
Extended Data Fig. 1x	N = 14 pairs of male mice GABAergic group and 13 pairs of male mice in glutamatergic group	Two-tailed Mann-Whitney test	Inter-brain PCC of GABAergic versus glutamatergic neurons: R1=273, R2=105, U=14, ****P < 0.0001
Extended Data Fig. 1y	N = 14 pairs of male mice in GABAergic across all groups.	Friedman test followed by Dunn's multiple comparison test	Q = 26.14, ****P < 0.0001
			Original vs Social: Z = 2.268, *P = 0.0467
			Original vs Non-social: Z = 2.835, **P = 0.0092
Extended Data Fig. 1z	N = 13 pairs of male mice in glutamatergic across all groups.	Friedman test followed by Dunn's multiple comparison test	Q = 4.769, P = 0.0921
			Original vs Social: Z = 1.373, P = 0.3396
			Original vs Non-social: Z = 0.7845, P = 0.8655
Extended Data Fig. 2a	N = 10 male mice in GABAergic N = 13 male mice in glutamatergic	Two-tailed Mann-Whitney test	Attack-Other: R1=185, R2=91, U=0, ****P < 0.0001
Extended Data Fig. 2b	N = 21 male mice in GABAergic N = 23 male mice in glutamatergic	Two-tailed Mann-Whitney test	Escape-Other: R1=639, R2=351, U=0, ****P < 0.0001
Extended Data Fig. 2c	N = 28 male mice in GABAergic N = 26 male mice in glutamatergic	Two-tailed Mann-Whitney test	Sniff-Other: R1=1102, R2=383, U=32, ****P < 0.0001
Extended Data Fig. 2d	N = 27 male mice in GABAergic N = 26 male mice in glutamatergic	Two-tailed Mann-Whitney test	Self groom-Other: R1=947, R2=484, U=133, ****P < 0.0001
Extended Data Fig. 2e	N = 27 male mice in GABAergic N = 23 male mice in glutamatergic	Two-tailed Mann-Whitney test	Dig-Other: R1=831, R2=444, U=168, **P = 0.0050
Extended Data Fig. 2f	N = 9 male mice in GABAergic N = 8 male mice in glutamatergic	Two-tailed Mann-Whitney test	Attack-Sniff: R1=117, R2=36, U=0, ****P < 0.0001
Extended Data Fig. 2g	N = 5 male mice in GABAergic N = 7 male mice in glutamatergic	Two-tailed Mann-Whitney test	Attack-Approach: R1=55, R2=36, U=0, **P = 0.0016

Extended Data Fig. 2h	N = 4 male mice in GABAergic N = 5 male mice in glutamatergic	Two-tailed Mann-Whitney test	Attack-Selfgroom: R1=74, R2=17, U=2, **P = 0.0062
Extended Data Fig. 2i	N = 27 male mice in GABAergic N = 23 male mice in glutamatergic	Two-tailed Mann-Whitney test	Sniff-Self groom: R1=890, R2=385, U=109, ****P < 0.0001
Extended Data Fig. 2j	N = 10 male mice in GABAergic N = 13 male mice in glutamatergic	Two-tailed Mann-Whitney test	Attack-Other: R1=179, R2=97, U=6, ****P < 0.0001
Extended Data Fig. 2k	N = 21 male mice in GABAergic N = 23 male mice in glutamatergic	Two-tailed Mann-Whitney test	Escape-Other: R1=634, R2=356, U=80, ****P < 0.0001
Extended Data Fig. 2l	N = 28 male mice in GABAergic N = 26 male mice in glutamatergic	Two-tailed Mann-Whitney test	Sniff-Other: R1=1086, R2=399, U=48, ****P < 0.0001
Extended Data Fig. 2m	N = 27 male mice in GABAergic N = 26 male mice in glutamatergic	Two-tailed Mann-Whitney test	Self groom-Other: R1=962, R2=469, U=118, ****P < 0.0001
Extended Data Fig. 2n	N = 27 male mice in GABAergic N = 23 male mice in glutamatergic	Two-tailed Mann-Whitney test	Dig-Other: R1=862, R2=413, U=137, ***P = 0.0005
Extended Data Fig. 3a	N = 14 pairs of male mice in GABAergic, N = 13 pairs of male mice in glutamatergic	Two-tailed Mann Whitney test	GABAergic-glutamatergic: R1=257, R2=121, U=30, **P = 0.0019
Extended Data Fig. 3b	N = 14 pairs of male mice in GABAergic	Two-tailed Wilcoxon matched-pairs signed rank test	GABAergic Interaction vs Separation: W = -105.0, ***P = 0.0001
Extended Data Fig. 3c	N = 13 pairs male mice in glutamatergic	Two-tailed Wilcoxon matched-pairs signed rank test	Glutamatergic Interaction vs Separation: W = -105.0, ***P = 0.0001
Extended Data Fig. 3d	N = 14 pairs of male mice in GABAergic	Two-tailed Wilcoxon matched-pairs signed rank test	GABAergic Interaction vs Separation: W = -11.0, P = 0.7609
Extended Data Fig. 3e	N = 13 pairs of male mice in glutamatergic	Two-tailed Wilcoxon matched-pairs signed rank test	Glutamatergic Interaction vs Separation: W = 37.0, P = 0.2676
Extended Data Fig. 3f	N = 28 male mice, 12 female mice in GABA. N = 22 male mice, 14 female mice in glutamatergic.	2-way ANOVA followed by Fisher's LSD test	Interaction: F(1,72) = 5.023, *P = 0.0281
			Cell type: F(1,72) = 109.0, ****P < 0.0001
			Sex: F(1,72) = 4.394, *P = 0.0396
			GABA male-female: t = 3.053, **P = 0.0032
			Glutamatergic male-female: t = 0.1030, P = 0.9183
			GABA male-Glutamatergic male: t = 10.81, ****P < 0.0001
Extended Data Fig. 3g	N = 14 pairs of male mice, 8 pairs of female mice in GABAergic; N = 13 pairs of male mice, 9 pairs of female mice in glutamatergic.	2-way ANOVA followed by Fisher's LSD test	GABA female-Glutamatergic female: t = 5.063, ****P < 0.0001
			Interaction: F(1,40) = 12.82, ***P = 0.0009
			Cell type: F(1,40) = 56.38, ****P < 0.0001
			Sex: F(1,40) = 13.98, ***P = 0.0006
			GABA male-female: t = 5.121, ****P < 0.0001
			Glutamatergic male-female: t = 0.1135, P = 0.9102
Extended Data Fig. 3h	N = 14 pairs of male mice, 8 pairs of female mice in GABAergic; N = 13 pairs of male mice, 9 pairs of female mice in glutamatergic.	2-way ANOVA followed by Fisher's LSD test	GABA male-Glutamatergic male: t = 8.926, ****P < 0.0001
			GABA female-Glutamatergic female: t = 2.506, *P = 0.0164
			Interaction: F(1,39) = 3.166, P = 0.083
			Cell type: F(1,39) = 1.011, P = 0.3209
			Sex: F(1,39) = 0.08812, P = 0.7582
			GABA male-female: t = 1.443, P = 0.1571
Extended Data Fig. 4a	N = 14 pairs of male mice in GABAergic Interaction and Separation.	Two-tailed Wilcoxon matched-pairs signed rank test	GABAergic Interaction vs Separation: W = -105.0, ***P = 0.0001
Extended Data Fig. 4b	N = 13 pairs of male mice in glutamatergic Interaction and Separation.	Two-tailed Wilcoxon matched-pairs signed rank test	glutamatergic Interaction vs Separation: W = -21.0, *P = 0.0312
Extended Data Fig. 4d	N = 14 pairs of male mice in GABAergic	Two-tailed Mann-Whitney test	GABAergic versus glutamatergic interaction R1=798, R2=105, U=0, ****P < 0.0001
Extended Data Fig. 4e	N = 13 pairs of male mice in glutamatergic	Two-tailed Mann-Whitney test	GABAergic versus glutamatergic interaction R1=279, R2=99, U=8, ****P < 0.0001
Extended Data Fig. 4f		Two-tailed Mann-Whitney test	GABAergic versus glutamatergic interaction R1=163, R2=188, U=72, P = 0.5446

Extended Data Fig. 4g	N = 14 pairs of male mice in GABAergic Interaction and Separation.	Two-tailed Wilcoxon matched-pairs signed rank test	GABAergic Interaction vs Separation: W = -105.0, ***P = 0.0001
Extended Data Fig. 4h	N = 13 pairs of male mice in glutamatergic Interaction and Separation.	Two-tailed Wilcoxon matched-pairs signed rank test	glutamatergic Interaction vs Separation: W = -55.0, **P = 0.0020
Extended Data Fig. 4j	N = 14 pairs of male mice in GABAergic N = 13 pairs of male mice in glutamatergic.	Two-tailed Mann-Whitney test	GABAergic versus glutamatergic interaction R1=1130, R2=355, U=4, ****P < 0.0001
Extended Data Fig. 4k		Two-tailed Mann-Whitney test	GABAergic versus glutamatergic interaction R1=287, R2=91, U=0, ****P < 0.0001
Extended Data Fig. 4l		Two-tailed Mann-Whitney test	GABAergic versus glutamatergic interaction R1=212, R2=166, U=75, P = 0.4583
Extended Data Fig. 4m	N = 14 pairs of male mice in GABAergic Interaction and Separation.	Two-tailed Wilcoxon matched-pairs signed rank test	GABAergic Interaction vs Separation: W = -105.0, ***P = 0.0001
Extended Data Fig. 4n	N = 13 pairs of male mice in glutamatergic Interaction and Separation.	Two-tailed Wilcoxon matched-pairs signed rank test	glutamatergic Interaction vs Separation: W = -36.0, **P = 0.0078
Extended Data Fig. 4p	N = 14 pairs of male mice in GABAergic N = 13 pairs of male mice in glutamatergic	Two-tailed Mann-Whitney test	GABAergic versus glutamatergic interaction R1=847, R2=143, U=7, ****P < 0.0001
Extended Data Fig. 4q		Two-tailed Mann-Whitney test	GABAergic versus glutamatergic interaction R1=216, R2=37, U=1, ****P < 0.0001
Extended Data Fig. 4r		Two-tailed Mann-Whitney test	GABAergic versus glutamatergic interaction R1=220, R2=158, U=67, P = 0.2588
Extended Data Fig. 4t	N = 14 pairs of male mice in GABAergic	Two-tailed Mann-Whitney test	GABAergic versus glutamatergic interaction R1=845, R2=145, U=9, ****P < 0.0001
Extended Data Fig. 4u	N = 13 pairs of male mice in glutamatergic.	Two-tailed Mann-Whitney test	GABAergic versus glutamatergic interaction R1=217, R2=36, U=0, ****P < 0.0001
Extended Data Fig. 4v	N = 14 pairs of male mice in GABAergic Interaction and Separation.	Two-tailed Wilcoxon matched-pairs signed rank test	GABAergic PSC1 vs CC1: W = -406.0, ****P < 0.0001
Extended Data Fig. 4w	N = 13 pairs of male mice in glutamatergic Interaction and Separation.	Two-tailed Wilcoxon matched-pairs signed rank test	glutamatergic PSC1 vs CC1: W = -136.0, ****P < 0.0001
Extended Data Fig. 5a	N = 14 male pairs in GABAergic across all groups.	One-way ANOVA, followed by Tukey's multiple comparisons test	F = 38.04, ****P < 0.0001
			Mutual-Unidirectional: Q = 5.832, *P = 0.032
			Mutual-Non-social: Q = 9.485, ****P < 0.0001
			Unidirectional-Non-social: Q = 16.05, ****P < 0.0001
Extended Data Fig. 5b		One-way ANOVA, followed by Tukey's multiple comparisons test	F = 11.31, **P = 0.0014
			Mutual-Unidirectional: Q = 3.505, P = 0.0669
			Mutual-Non-social: Q = 5.70, **P = 0.0038
			Unidirectional-Non-social: Q = 4.531, *P = 0.0177
Extended Data Fig. 5c		One-way ANOVA, followed by Tukey's multiple comparisons test	F = 17.69, ****P < 0.0001
			Mutual-Unidirectional: Q = 5.423, **P = 0.0055
			Mutual-Non-social: Q = 8.245, ****P = 0.0002
			Unidirectional-Non-social: Q = 2.251, P = 0.2838
Extended Data Fig. 5d		One-way ANOVA, followed by Tukey's multiple comparisons test	F = 21.72, ****P < 0.0001
			Mutual-Unidirectional: Q = 3.898, *P = 0.0405
			Mutual-Non-social: Q = 7.955, ****P = 0.0002
			Unidirectional-Non-social: Q = 7.131, ***P = 0.0006
Extended Data Fig. 5e	N = 13 male pairs in glutamatergic across all groups.	One-way ANOVA, followed by Tukey's multiple comparisons test	F = 9.588, **P = 0.0032
			Mutual-Unidirectional: Q = 2.009, P = 0.3616
			Mutual-Non-social: Q = 5.219, **P = 0.0080
			Unidirectional-Non-social: Q = 3.990, *P = 0.0381
Extended Data Fig. 5f		One-way ANOVA, followed by Tukey's multiple comparisons test	F = 7.552, **P = 0.0073
			Mutual-Unidirectional: Q = 2.004, P = 0.3633
			Mutual-Non-social: Q = 5.477, **P = 0.0058
			Unidirectional-Non-social: Q = 4.341, *P = 0.0245
Extended Data Fig. 5g		One-way ANOVA, followed by Tukey's multiple comparisons test	F = 4.042, *P = 0.0444
			Mutual-Unidirectional: Q = 2.751, P = 0.1687
			Mutual-Non-social: Q = 3.250, P = 0.0946
			Unidirectional-Non-social: Q = 2.028, P = 0.3554
Extended Data Fig. 5h		One-way ANOVA, followed by Tukey's multiple comparisons test	F = 20.81, ****P < 0.0001
			Mutual-Unidirectional: Q = 3.681, P = 0.0561
			Mutual-Non-social: Q = 9.272, ****P < 0.0001
			Unidirectional-Non-social: Q = 5.852, **P = 0.0036

Extended Data Fig. 5i		Two-tailed Mann-Whitney test	GABAergic mutual social-rearing: U=17, R1=214, R2=62, **P = 0.0027 Glutamatergic mutual social-rearing: U=35, R1=173, R2=80, P = 0.1264
Extended Data Fig. 5j	GABAergic mutual social: N = 14 male mice GABAergic rearing: N = 9 male mice	Two-tailed Mann-Whitney test	GABAergic mutual social-rearing: U=8, R1=223, R2=53, ***P = 0.0002 Glutamatergic mutual social-rearing: U=57, R1=151, R2=102, P = 0.09479
Extended Data Fig. 5k	Glutamatergic mutual social: N = 13 male mice Glutamatergic rearing: N = 9 male mice	Two-tailed Mann-Whitney test	GABAergic mutual social-rearing: U=11, R1=220, R2=56, ***P = 0.0005 Glutamatergic mutual social-rearing: U=49, R1=159, R2=94, P = 0.5556
Extended Data Fig. 5l		Two-tailed Mann-Whitney test	GABAergic mutual social-rearing: U=21.5, R1=209.5, R2=66.5, **P = 0.007 Glutamatergic mutual social-rearing: U=27.5, R1=154.5, R2=55.5, P = 0.1623
Extended Data Fig. 5m	N = 26 male mice in glutamatergic	Two-tailed Mann-Whitney test	U=35, R1=211, R2=140, *P = 0.0103
Extended Data Fig. 6a	N = 14 pairs of male mice in GABAergic Interaction and Separation.	Two-tailed Wilcoxon matched-pairs signed rank test	GABAergic Interaction vs Separation: W = -105.0, ***P = 0.0001
Extended Data Fig. 6b	N = 13 pairs of male mice in glutamatergic Interaction and Separation.	Two-tailed Wilcoxon matched-pairs signed rank test	Glutamatergic Interaction vs Separation: W = -66.0, ***P = 0.0010
Extended Data Fig. 6c	N = 14 pairs of male mice in GABAergic N = 13 pairs of male mice in glutamatergic.	Two-tailed Mann-Whitney test	GABAergic versus glutamatergic interaction R1=1116, R2=369, U=18, ****P < 0.0001
Extended Data Fig. 6d	PLSC1: N = 18 male mice in GABAergic Interaction U1: N = 28 male mice in GABAergic Interaction	Two-tailed Wilcoxon matched-pairs signed rank test Two-tailed Wilcoxon matched-pairs signed rank test	GABAergic Observed vs Permuted: W = -155.0, ***P = 0.0002 GABAergic Observed vs Permuted: W = 42.0, P = 0.6456
Extended Data Fig. 6e	PLSC1: N = 12 male mice in glutamatergic Interaction U1: N = 26 male mice in glutamatergic Interaction	Two-tailed Wilcoxon matched-pairs signed rank test Two-tailed Wilcoxon matched-pairs signed rank test	glutamatergic Observed vs Permuted: W = -54.0, *P = 0.0342 glutamatergic Observed vs Permuted: W = -3.000, P = 0.9801
Extended Data Fig. 6f	N = 24 male mice in GABAergic Interaction	Two-tailed Wilcoxon matched-pairs signed rank test	GABAergic Observed vs Permuted: W = 126.0, P = 0.0738
Extended Data Fig. 6g		Two-tailed Wilcoxon matched-pairs signed rank test	GABAergic Observed vs Permuted: W = 242.0, ***P = 0.0002
Extended Data Fig. 6h	N = 24 male mice in Glutamatergic Interaction	Two-tailed Wilcoxon matched-pairs signed rank test	GABAergic Observed vs Permuted: W = 224.0, ***P = 0.0007
Extended Data Fig. 6i		Two-tailed Wilcoxon matched-pairs signed rank test	GABAergic Observed vs Permuted: W = 6.000, P = 0.9441
Extended Data Fig. 6j	Attack: N = 9, Defend: N = 21, Sniff: N = 28, Non-social: N = 28, Approach: N = 22 male mice in GABAergic	Wilcoxon matched-pairs signed rank test	Attack: W = -45.0, **P = 0.0039; Defend: W = -207.0, ****P < 0.0001; Sniff: W = -314.0, ***P = 0.0001; Non-social: W = -398.0, ****P < 0.0001; Approach: W = -243.0, ****P < 0.0001
Extended Data Fig. 6k	Attack: N = 12, Defend: N = 25, Sniff: N = 25, Non-social: N = 26, Approach: N = 25 male mice in glutamatergic	Wilcoxon matched-pairs signed rank test	Attack: W = -76.0, ***P = 0.001; Defend: W = -231.0, **P = 0.0012; Sniff: W = 119.0, P = 0.1135; Non-social: W = -285.0, ****P < 0.0001; Approach: W = -247.0, ***P = 0.0004
Extended Data Fig. 6l	PLSC1: N = 28 male mice in GABAergic across all groups	Kruskal Wallis test followed by Dunn's multiple comparison test	H=54.59, P < 0.0001 State+Transit-State: Z = 1.035, P = 0.3005 State+Transit-Delta: Z = 6.853, P < 0.0001 State-Delta: Z = 5.818, P < 0.0001
			H=19.87, P < 0.0001 State+Transit-State: Z = 2.383, P = 0.0172 State+Transit-Delta: Z = 4.454, P < 0.0001 State-Delta: Z = 2.071, P = 0.0384
			H=22.81, P < 0.0001 State+Transit-State: Z = 2.534, P = 0.0113 State+Transit-Delta: Z = 4.773, P < 0.0001 State-Delta: Z = 2.240, P = 0.0251
Extended Data Fig. 6m	PLSC1: N = 26 male mice in glutamatergic across all groups	Kruskal Wallis test followed by Dunn's multiple comparison test	

	U1: N = 26 male mice in glutamatergic across all groups	Kruskal Wallis test followed by Dunn's multiple comparison test	H=23.33, P < 0.0001 State+Transit-State: Z = 2.729, P = 0.0063 State+Transit-Delta: Z = 4.816, P < 0.0001 State-Delta: Z = 2.087, P = 0.0369
Extended Data Fig. 6n	N = 28 male mice in GABAergic	Two-tailed Wilcoxon matched-pairs signed rank test	W = 142.0, P = 0.1093
Extended Data Fig. 6o		Two-tailed Wilcoxon matched-pairs signed rank test	W = -100.0, P = 0.2642
Extended Data Fig. 6p	N = 22 male mice in Glutamatergic	Two-tailed Wilcoxon matched-pairs signed rank test	W = -133.0, *P = 0.0301
Extended Data Fig. 6q		Two-tailed Wilcoxon matched-pairs signed rank test	W = -37.00, P = 0.5661
Extended Data Fig. 7a	N = 6 pairs of GABAergic experiments	Wilcoxon matched-pairs signed rank test	W = -21.0, *P = 0.0312
Extended Data Fig. 7b	N = 4 pairs of glutamatergic experiments	Wilcoxon matched-pairs signed rank test	W = -10.0, P = 0.1250
Extended Data Fig. 7c	Attack-Defend: N = 10, Attack-Sniff: N = 12, Defend-Sniff: N = 14, Attack-Approach: N = 12, Defend-Approach: N = 13, Sniff-Non-social: N = 14,	Wilcoxon matched-pairs signed rank test	Attack-Defend: W = -43.0, *P = 0.0273 Attack-Sniff: W = -58.0, *P = 0.0210, Defend-Sniff: W = -99.0, ***P = 0.0006, Attack-Approach: W = -68.0, ***P = 0.0049, Defend-Approach: W = -81.0, **P = 0.0024, Sniff-Non-social: W = -103.0, ***P = 0.0002, Approach-Non-social: W = -103.0, ***P = 0.0002, Sniff-Approach: W = -101.0, ***P = 0.0004, Attack-Non-social: W = -70.0, **P = 0.0034, Defend-Non-social: W = -99.0, ***P = 0.0006
Extended Data Fig. 7d	Approach-Non-social: N = 14, Sniff-Approach: N = 14, Attack-Non-social: N = 12, Defend-Non-social: N = 14 pairs of male mice in GABAergic	Wilcoxon matched-pairs signed rank test	Attack-Defend: W = -17.0, P = 0.4316 Attack-Sniff: W = -16.0, P = 0.5693, Defend-Sniff: W = -33.0, P = 0.3258, Attack-Approach: W = -22.0, P = 0.4238, Defend-Approach: W = -17.0, P = 0.5879, Sniff-Non-social: W = 15.0, P = 0.6698, Approach-Non-social: W = -37.0, P = 0.2676, Sniff-Approach: W = -7.0, P = 0.8552, Attack-Non-social: W = -38.0, P = 0.1514, Defend-Non-social: W = 3.0, P = 0.9515
Extended Data Fig. 7e	Attack-Defend: N = 10, Attack-Sniff: N = 10, Defend-Sniff: N = 13, Attack-Approach: N = 10, Defend-Approach: N = 13, Sniff-Non-social: N = 13,	Wilcoxon matched-pairs signed rank test	Attack-Defend: W = -29.0, P = 0.1602 Attack-Sniff: W = -35.0, P = 0.0840, Defend-Sniff: W = -35.0, P = 0.2439, Attack-Approach: W = -49.0, **P = 0.0098, Defend-Approach: W = -83.0, **P = 0.0017, Sniff-Non-social: W = -85.0, **P = 0.0012, Approach-Non-social: W = -91.0, ***P = 0.0002, Sniff-Approach: W = -89.0, ***P = 0.0005, Attack-Non-social: W = -43.0, *P = 0.0273, Defend-Non-social: W = -87.0, ***P = 0.0007
Extended Data Fig. 7f	Approach-Non-social: N = 13, Sniff-Approach: N = 13, Attack-Non-social: N = 10, Defend-Non-social: N = 13 pairs of male mice in glutamatergic	Wilcoxon matched-pairs signed rank test	Attack-Defend: W = -17.0, P = 0.4316 Attack-Sniff: W = 11.0, P = 0.6250, Defend-Sniff: W = -59.0, *P = 0.0398, Attack-Approach: W = -31.0, P = 0.1309, Defend-Approach: W = -51.0, P = 0.0803, Sniff-Non-social: W = -73.0, **P = 0.0081, Approach-Non-social: W = -103.0, P = 0.8926, Sniff-Approach: W = -17.0, P = 0.5879, Attack-Non-social: W = -13.0, P = 0.5566, Defend-Non-social: W = -7.0, P = 0.8394
Extended Data Fig. 7h	Attack: Escape-Defend: N = 9; Defend: Attack-Sniff: N = 11; Attack-Approach: N = 4, Sniff-Approach: N = 6; Sniff: Defend-Sniff: N = 18, Defend-Non-social: N = 12, Sniff-Non-social: N = 18; Non-social: Sniff-Approach: N = 13, Sniff-Non-social: N = 13,	Wilcoxon matched-pairs signed rank test	Attack: Escape-Defend: W = -45.0, **P = 0.0039; Defend: Attack-Sniff: W = -50.0, *P = 0.0244 Attack-Approach: W = -10.0, P = 0.1250; Sniff-Approach: W = -13.0, P = 0.2188; Sniff: Defend-Sniff: W = -89.0, P = 0.0539, Defend-Non-social: W = 30.0, P = 0.2611, Sniff-Non-social: W = 21.0, P = 0.6705; Non-social: Sniff-Approach: W = -47.0, P = 0.1099, Sniff-Non-social: W = -131.0, **P = 0.0028

	Approach-Non-social: N = 10 male mice in GABAergic		Approach-Non-social: W = -108.0, **P = 0.0034,
Extended Data Fig. 7i	Attack: Attack-Escape: N = 6, Attack-Defend: N = 6, Escape-Defend: N = 7; Defend: Attack-Sniff: N = 13, Attack-Approach: N = 12, Sniff-Approach: N = 14; Sniff: Defend-Sniff: N = 21, Defend-Non-social: N = 14, Sniff-Non-social: N = 14; Non-social: Sniff-Approach: N = 16, Sniff-Non-social: N = 16, Approach-Non-social: N = 22 male mice in glutamatergic	Wilcoxon matched-pairs signed rank test	Attack: Attack-Escape: W = -11.0, P = 0.3125, Attack-Defend: W = -11.0, P = 0.3125, Escape-Defend: W = -26.0, *P = 0.0312; Defend: Attack-Sniff: W = -25.0, P = 0.4143 Attack-Approach: W = -44.0, P = 0.0923; Sniff-Approach: W = -55.0, P = 0.0906; Sniff: Defend-Sniff: W = -95.0, P = 0.1032, Defend-Non-social: W = -23.0, P = 0.5016, Sniff-Non-social: W = -27.0, P = 0.4263; Non-social: Sniff-Approach: W = -22.0, P = 0.5966, Sniff-Non-social: W = -36.0, P = 0.3755 Approach-Non-social: W = 53.0, P = 0.4060,
Extended Data Fig. 8a	N = 22 male mice in glutamatergic	Friedman test followed by Dunn's multiple comparison test	Friedman statistic 71.61, ****p < 0.0001 Attack vs Escape: Z = 0.1412, P > 0.9999 Attack vs Climb: Z = 0.2824, P > 0.9999 Attack vs Stand: Z = 0.1694, P > 0.9999 Attack vs Chasing: Z = 1.355, P > 0.9999 Attack vs Dig: Z = 0.09884, P > 0.9999 Attack vs Defend: Z = 0.6495, P > 0.9999 Attack vs General-sniffing: Z = 0.5930, P > 0.9999 Attack vs Tussling: Z = 2.810, P = 0.0843 Attack vs Follow: Z = 1.158, P > 0.9999 Attack vs Selfgrooming: Z = 0.9601, P > 0.9999 Attack vs Approach: Z = 1.158, P > 0.9999 Attack vs Sniff-face: Z = 1.638, P > 0.9999 Attack vs Sniff-genital: Z = 3.163, *P = 0.0266 Attack vs Explore object: Z = 2.697, P = 0.1190 Attack vs Threaten: Z = 3.968, **P = 0.0012 Attack vs Flinch: Z = 3.163, *P = 0.0266 Attack vs Bite object: Z = 3.812, **P = 0.0023
Extended Data Fig. 8c	N = 7 pairs of male mice in GABAergic	Two-tailed Wilcoxon matched-pairs signed rank test	Offensive: W = -28.00, *P = 0.0156
Extended Data Fig. 8d		Two-tailed Wilcoxon matched-pairs signed rank test	Defensive: W = 26.00, *P = 0.0312
Extended Data Fig. 8e		Two-tailed Wilcoxon matched-pairs signed rank test	Investigative: W = -22.00, P = 0.0781
Extended Data Fig. 8f		Kruskal-Wallis test followed by Dunn's multiple comparison test	H=16.76, ***P = 0.0008 Interaction obs-Interaction shuffle: Z = 3.606, ***P = 0.0003 Separation obs-Separation shuffle: Z = 1.007, P = 0.3138 Interaction obs-Separation obs: Z = 2.469, *P = 0.0135
Extended Data Fig. 8g	N = 6 pairs of male mice in glutamatergic	Kruskal-Wallis test followed by Dunn's multiple comparison test	H=10.87, *P = 0.0124 Interaction obs-Interaction shuffle: Z = 3.021, **P = 0.0025 Separation obs-Separation shuffle: Z = 0.8165, P = 0.4142 Interaction obs-Separation obs: Z = 1.837, P = 0.0662
Extended Data Fig. 8h	N = 7 pairs of male mice in GABAergic	Wilcoxon matched-pairs signed rank test	W = 28.0, *P = 0.0156
Extended Data Fig. 8i	N = 6 pairs of male mice in glutamatergic	Wilcoxon matched-pairs signed rank test	W = 21.0, *P = 0.0312
Extended Data Fig. 8j	N = 7 pairs of male mice in GABAergic; N = 6 pairs of male mice in glutamatergic	Two-tailed Mann-Whitney test	U=3, ****P < 0.0001
Extended Data Fig. 8k		Two-tailed Mann-Whitney test	U=0, **P = 0.0012
Extended Data Fig. 8l		Two-tailed Mann-Whitney test	U=21, P > 0.9999
Extended Data Fig. 8m	N = 7 male mice in GABAergic across all groups	Friedman test followed by Dunn's multiple comparison test	H=12.29, ***P = 0.0003 Aggression-Non-aggression: Z = 1.336, P = 0.1824 Aggression-Non-social: Z = 3.474, ***P = 0.0005 Non-aggression-Non-social: Z = 2.138, *P = 0.0325
Extended Data Fig. 8n	N = 6 male mice in glutamatergic across all groups		H=4.333, P = 0.1416 Aggression-Non-aggression: Z = 0.5774, P = 0.5637

		Friedman test followed by Dunn's multiple comparison test	Aggression-Non-social: Z = 2.021, *P = 0.0433 Non-aggression-Non-social: Z = 1.443, P = 0.1489
Extended Data Fig. 8o	N = 7 male mice in GABAergic across all groups	Friedman test followed by Dunn's multiple comparison test	H=12.29, ***P = 0.0003 Aggression-Non-aggression: Z = 2.138, *P = 0.0325 Aggression-Non-social: Z = 3.474, ***P = 0.0005 Non-aggression-Non-social: Z = 1.336, P = 0.1814
Extended Data Fig. 8p	N = 6 male mice in glutamatergic across all groups	Friedman test followed by Dunn's multiple comparison test	H=6.333, P = 0.0521 Aggression-Non-aggression: Z = 2.021, *P = 0.0433 Aggression-Non-social: Z = 2.309, *P = 0.0209 Non-aggression-Non-social: Z = 0.2887, P = 0.7728
Extended Data Fig. 9a	N = 26 male mice in glutamatergic	Friedman test followed by Dunn's multiple comparison test	Friedman statistic = 148.7, ****P < 0.0001 Social versus Social Permuted: Z = 5.435, ****P < 0.0001 Non-social versus Non-social Permuted: Z = 4.869, ****P < 0.0001 Separation versus Separation Permuted: Z = 4.699, ****P < 0.0001 Social versus Non-social: Z = 0.1698, P > 0.9999 Social versus Separation: Z = 1.755, P = 0.5548 Non-social versus Separation: Z = 1.925, P = 0.4340 Interaction versus Interaction Permuted: Z = 5.152, ****P < 0.0001 Interaction versus Separation: Z = 4.586, ****P < 0.0001
Extended Data Fig. 9b		Friedman test followed by Dunn's multiple comparison test	Friedman statistic = 122.9, ****P < 0.0001 Social versus Social Permuted: Z = 7.586, ****P < 0.0001 Non-social versus Non-social Permuted: Z = 5.095, ****P < 0.0001 Separation versus Separation Permuted: Z = 2.831, *P = 0.0372 Social versus Non-social: Z = 2.208, P = 0.2180 Social versus Separation: Z = 0.8492, P > 0.9999 Non-social versus Separation: Z = 0.8492, P > 0.9999 Interaction versus Interaction Permuted: Z = 4.303, ****P = 0.0001 Interaction versus Separation: Z = 0.9058, P > 0.9999
Extended Data Fig. 9c	N = 13 pairs of male mice in glutamatergic	Linear regression	F = 0.05982, P = 0.8123
Extended Data Fig. 9d		Linear regression	F = 0.01657, P = 0.8999
Extended Data Fig. 9e		Linear regression	F = 4.650, P = 0.0594
Extended Data Fig. 9f		Linear regression	F = 0.7544, P = 0.4036
Extended Data Fig. 9g	N = 22929 cell pairs	Two-tailed Wilcoxon matched-pairs signed rank test	W = -111222471, ****P < 0.0001
Extended Data Fig. 9h	N = 50673 cell pairs	Two-tailed Wilcoxon matched-pairs signed rank test	W = 469449125, ****P < 0.0001
Extended Data Fig. 9j	N = 28 male mice in GABAergic	Friedman test followed by Dunn's multiple comparison test	Friedman statistic = 123.5, ****P < 0.0001 Social versus Social Permuted: Z = 6.857, ****P < 0.0001 Non-social versus Non-social Permuted: Z = 5.929, ****P < 0.0001 Separation versus Separation Permuted: Z = 2.831, ****P < 0.0001 Social versus Non-social: Z = 2.429, P = 0.0910 Social versus Separation: Z = 3.357, **P = 0.0047 Non-social versus Separation: Z = 0.9286, P = 0.0001
Extended Data Fig. 9k		Friedman test followed by Dunn's multiple comparison test	Friedman statistic = 122.6, ****P < 0.0001 Social versus Social Permuted: Z = 6.786, ****P < 0.0001 Non-social versus Non-social Permuted: Z = 7.857, ****P < 0.0001 Separation versus Separation Permuted: Z = 3.357, **P = 0.0047 Social versus Non-social: Z = 0.7857, P > 0.9999 Social versus Separation: Z = 2.214, P = 0.1609 Non-social versus Separation: Z = 1.429, P = 0.9188
Extended Data Fig. 9l	N = 28 male mice in glutamatergic	Friedman test followed by Dunn's multiple comparison test	Q = 15.57, ***P = 0.0004 Original-xIntrabrain: Z = 3.591, ***P = 0.0007 Original-xTemporal coupling: Z = 0.000, P > 0.9999

Extended Data Fig. 9m	N = 28 males in GABAergic and 26 males in glutamatergic	Two-tailed Mann-Whitney test	GABAergic versus glutamatergic interaction: R1=972, R2=513, U=162, ***P = 0.0003
Extended Data Fig. 9n	N = 28 males in GABAergic and 26 males in glutamatergic	Two-tailed Mann-Whitney test	GABAergic versus glutamatergic separation: R1=1012, R2=473, U=162, ***P < 0.0001
Extended Data Fig. 9o	N = 28 males in GABAergic and 26 males in glutamatergic	Two-tailed Mann-Whitney test	GABAergic versus glutamatergic interaction: R1=1103, R2=382, U=31, ****P < 0.0001
Extended Data Fig. 9p	N = 28 males in GABAergic and 25 males in glutamatergic	Two-tailed Mann-Whitney test	GABAergic versus glutamatergic separation: R1=1080, R2=351.5, U=26.5, ****P < 0.0001
Extended Data Fig. 9q	N = 17867 cell pairs in glutamatergic	Two-tailed Wilcoxon matched-pairs signed rank test	W = -97340736, ****P < 0.0001
Extended Data Fig. 9r	N = 10864 cell pairs in glutamatergic	Two-tailed Wilcoxon matched-pairs signed rank test	W = 33612070, ****P < 0.0001
Extended Data Fig. 9s	N = 8950 cell pairs in GABAergic and 17867 cell pairs in glutamatergic	Two-tailed Mann-Whitney test	GABAergic versus glutamatergic: R1=149868559, R2=209720594, U=50096816, ****P < 0.0001
Extended Data Fig. 9t	N = 7579 cell pairs in GABAergic and 10864 cell pairs in glutamatergic	Two-tailed Mann-Whitney test	GABAergic versus glutamatergic: R1=76880519, R2=93200827, U=34182147, ****P < 0.0001
Extended Data Fig. 10b	N = 24 male mice	Two-tailed Wilcoxon matched-pairs signed rank test	Attack: Shuffled-Observation W = -300, ****P < 0.0001
	N = 13 male mice		Chase: Shuffled-Observation W = -91, ***P = 0.0002
	N = 47 male mice		Escape: Shuffled-Observation W = -1128, ****P < 0.0001
	N = 44 male mice		Defend: Shuffled-Observation W = -984, ****P < 0.0001
	N = 54 male mice		Sniff: Shuffled-Observation W = -1485, ****P < 0.0001
	N = 50 male mice		Approach: Shuffled-Observation W = -1275, ****P < 0.0001
	N = 54 male mice		Selfgroom: Shuffled-Observation W = -1485, ****P < 0.0001
	N = 44 male mice		Dig: Shuffled-Observation W = -990, ****P < 0.0001
	N = 45 male mice		Stand: Shuffled-Observation W = -1035, ****P < 0.0001
Extended Data Fig. 10c	N = 28 male mice	Direct comparison with a null distribution	In these figures, the fraction of neural variance explained is presented as the value exceeding chance levels (determined from a null distribution).
Extended Data Fig. 10d	N = 26 male mice		
Extended Data Fig. 11b	N = 54 male mice		No statistical comparisons
Extended Data Fig. 11c	N = 28 male mice		No statistical comparisons
Extended Data Fig. 11d	N = 26 male mice		No statistical comparisons
Extended Data Fig. 11e	N = 12 behaviors	Linear Regression	F = 0.4080, P = 0.5373
Extended Data Fig. 11f		Linear Regression	F = 0.5743, P = 0.4660
Extended Data Fig. 11h	N = 7 pairs of male mice	Two-tailed Wilcoxon matched-pairs signed rank test	W = 8.00, P = 0.5781
Extended Data Fig. 11j		Two-tailed Wilcoxon matched-pairs signed rank test	W = 0.00, P > 0.9999
Extended Data Fig. 11l		Two-tailed Wilcoxon matched-pairs signed rank test	W = -8.00, P = 0.5781
Extended Data Fig. 12a	N = 22 across all groups	Friedman test and Dunn's multiple comparisons test	Q = 44.00, ****P < 0.0001
			Shared vs Unique: Z = 6.633, ****P < 0.0001
			Shared vs Full: Z = 3.317, **P = 0.0027
			Unique vs Full: Z = 3.317, **P = 0.0027
Extended Data Fig. 12b	N = 22 across all groups		No statistical comparisons
Extended Data Fig. 12c			
Extended Data Fig. 12d			
Extended Data Fig. 12e			
Extended Data Fig. 12f	N = 28 male mice	Two-tailed Wilcoxon matched-pairs signed rank test	W = 330.0, ****P < 0.0001
Extended Data Fig. 12g	N = 22 male mice	Two-tailed Wilcoxon matched-pairs signed rank test	W = 131.0, *P = 0.0329
Extended Data Fig. 13a	N = 56 in Social agents N = 58 Non-social agents with full vision	Kruskal-Wallis test, followed by Dunn multiple comparison test (part of Fig 6g)	Kruskal-Wallis H=153.9 ****P < 0.0001
			Social-Non-social with full vision: Z = 8.538, ****P < 0.0001
			Kruskal-Wallis H=167.2, ****P < 0.0001

Extended Data Fig. 13b		Kruskal-Wallis test, followed by Dunn's multiple comparison test (part of Fig 6h)	Social-Non-social with full vision: $Z = 6.402$, **** $P < 0.0001$
Extended Data Fig. 13c	N = 56 in Social agents	Two-tailed Wilcoxon matched-pairs signed rank test	$W = -1584$, **** $P < 0.0001$
Extended Data Fig. 13d		Two-tailed Wilcoxon matched-pairs signed rank test	$W = -760$, ** $P = 0.0016$
Extended Data Fig. 13e	N = 59 pairs of Social agents	Friedman test followed by Dunn's multiple comparison tests	$Q = 89.53$, **** $P < 0.0001$
			Original-xIntraset: $Z = 8.653$, **** $P < 0.0001$
			Original-xTemporal Coupling: $Z = 1.013$, $P = 0.6225$
Extended Data Fig. 13f	N = 33 pairs of agents	Two-tailed Wilcoxon matched-pairs signed rank test	$W = 549.0$, **** $P < 0.0001$
Extended Data Fig. 13g	N = 10 Social chasers	Two-tailed Wilcoxon matched-pairs signed rank test	$W = -37.00$, $P = 0.0645$
Extended Data Fig. 13i	N = 4 pairs of agents for each group of trained and untrained agents	Two-tailed Mann-Whitney test	Untrained versus trained agents: $R1=10$, $R2=26$, $U=0$, * $P = 0.0286$
Extended Data Fig. 13j		Two-tailed Mann-Whitney test	Untrained versus trained agents: $R1=10$, $R2=26$, $U=0$, * $P = 0.0286$
Extended Data Fig. 13k	N = 24 pairs of agents in collaborative, 56 pairs of agents in competitive, and 59 pairs of agents in non-social	Kruskal-Wallis test followed by Dunn's multiple comparison tests	**** $P < 0.0001$, $H=123.8$
			Collaborative-Competitive: **** $P < 0.0001$, $Z = 4.176$
			Collaborate-Non-social: **** $P < 0.0001$, $Z = 10.26$
		Kruskal-Wallis test followed by Dunn's multiple comparison tests	Competitive-Non-social: **** $P < 0.0001$, $Z = 7.850$
			**** $P < 0.0001$, $H=123.8$
			Collaborative-Competitive: ** $P = 0.0017$, $Z = 3.441$
			Collaborate-Non-social: **** $P < 0.0001$, $Z = 9.556$
			Competitive-Non-social: **** $P < 0.0001$, $Z = 7.901$
Extended Data Fig. 13n	N = 16 agents per group of trained and untrained agents	Two-tailed Mann-Whitney test	Untrained versus trained agents; $R1=166$, $R2=362$, $U=30$, **** $P < 0.0001$
Extended Data Fig. 13o	N = 8 pairs of agents per group	Two-tailed Wilcoxon matched-pairs signed rank test	$W = -28.00$, * $P = 0.0156$
Extended Data Fig. 13q	N = 25 agents per group of trained and untrained agents	Two-tailed Mann-Whitney test	Untrained versus trained agents; $R1=598$, $R2=677$, $U=273$, $P = 0.4496$
Extended Data Fig. 13r		Two-tailed Mann-Whitney test	Untrained versus trained agents; $R1=325$, $R2=950$, $U=0$, **** $P < 0.0001$
Extended Data Fig. 13s		Two-tailed Mann-Whitney test	Untrained versus trained agents; $R1=325.5$, $R2=949.5$, $U=0.5000$, **** $P < 0.0001$
Extended Data Fig. 13t	N = 25 pairs of agents	Two-tailed Wilcoxon matched-pairs signed rank test	$W = -325$, **** $P < 0.0001$

Supplementary Table 2. Raw and derived behavioral features

A total of 75 features were derived from the xy-position of the 7 nodes. “+” indicates the use of a feature in the behavior data while “-” indicates otherwise.

	Raw value	First derivative /Absolute value	Second derivative /Absolute value
Centroid Position	-	+/-	+/+
Distance from corner of cage to scope base	-	+/+	+/+
Distance from nose to scope base	+	+/+	+/+
Distance from body node to scope base	+	+/+	+/+
Distance from body node to tail	+	+/+	+/+
Absolute facing angle	Sine & Cosine	-/+	+/+
Absolute angle of scope base velocity vector	Sine & Cosine	-/+	+/+
Absolute angle of centroid velocity vector	Sine & Cosine	-/+	+/+
Body bend	+	+/+	+/+
Nose bend	+	+/+	+/+
Body area	+	+/+	+/+
Body area-Right	+	+/+	+/+
Body area-Left	+	+/+	+/+
Head pitch	+	-/+	+/+
Head roll	+	-/+	+/+
Head yaw	Sine & Cosine	-/+	+/+

Supplementary Table 3. Rewards used in training social and non-social agents

	Social		Non-social	
	Chaser	Explorer	Chaser	Explorer
Collision	+1.0	-1.0	-0.1	-0.5
New Fields Explored	+0.1	+1.0	+0.1	+1.0
Otherwise	-0.1	-0.5	-0.1	-0.5

Supplementary Table 4. Rewards used in training an alternative set of social agents in Extended Data Fig. 13f

	Chaser	Explorer
Collision	+1.0	-1.0
New Fields Explored	+0.1	+0.1
Otherwise	-0.1	-0.1

Supplementary Table 5. Rewards used in the coins task in Extended Data Fig. 13m

Agent 1/ Agent 2	No Coin	Own Coin	Partner's Coin
No Coin	+0/+0	+0/+1	-2/+1
Own Coin	+1/+0	+1/+1	-1/+1
Partner's Coin	+1/-2	+1/-1	-1/-1

Supplementary Table 6. Rewards used in a complex social task in Extended Data Fig. 13p

	Gatherer	Capturer
Apple	+1.0	0
Eat 1/3 acorn	+6.0	0
Capture	0	+1.0

Supplementary Notes

Supplementary Note 1

In all our analyses, the neural activity of each cell is z-scored, such that all neurons (both GABAergic and glutamatergic) are normalized to have the same activity mean and standard deviation. We found that GABAergic neurons exhibited higher firing rates than glutamatergic neurons (Extended Data Fig. 1t). To examine whether differences in firing rates might contribute to the observed inter-brain correlation differences between GABAergic and glutamatergic populations, we subsampled the top 20% of glutamatergic neurons with the highest firing rates, aligning their average firing rates more closely with those of GABAergic neurons. We found that the inter-brain correlation of GABAergic neurons remained higher than that of selected glutamatergic neurons (Extended Data Fig. 1u, v). As an alternative approach, we also subsampled the top 50% of glutamatergic neurons and the bottom 50% of GABAergic neurons based on average firing rates to achieve more similar firing rates. Similarly, we found that the inter-brain correlation of this selected

GABAergic population was higher than that of the corresponding glutamatergic population (Extended Data Fig. 1w, x). Together, these results suggest that differences in activity levels between neuron subtypes do not account for the observed inter-brain correlation differences across cell types.

Supplementary Note 2

We computed the PCC considering only neurons that responded to social or nonsocial behaviors (Extended Data Fig. 1y, z). In GABAergic neurons, the PCC of social-responsive cells is significantly higher than that of the full population, whereas the PCC of non-social cells is lower. In contrast, in glutamatergic neurons, the PCC of social-responsive cells and non-social-responsive cells is not significantly different from the full population. This further supports the conclusion that high inter-brain correlation emerges from social encoding cells in GABAergic neurons.

Supplementary Note 3

In addition to PLSC, we also used an alternative method, canonical correlation analysis (CCA), to examine the shared neural subspace. We found that the top shared dimensions identified using CCA were highly correlated with those identified using PLSC (Extended Data Fig. 4s). Additionally, in the CCA analysis, the top neural dimension captured approximately 3.5 times more neural variance in the GABAergic subpopulation than in the glutamatergic population (Extended Data Fig. 4t). Thus, our conclusions that shared neural dimensions arise during social interactions and that the GABAergic subpopulation contains a larger shared neural subspace than glutamatergic neurons are consistent between the two methods (Extended Data Fig. 4s-u). Nevertheless, it is important to note that we do not expect the exact shared subspace to be identical between the two methods, as they are mathematically different—CCA maximizes projection correlation, whereas PLSC maximizes projection covariance. CCA may therefore identify dimensions that have correlated activity, but account for only a small fraction of the total variance, making it more susceptible to noise (Extended Data Fig. 4v, w).

Supplementary Note 4

The social interaction sessions (20 mins) were longer than the separation sessions (10 mins). To rule out the possibility that the absence of shared neural dimensions in the separation session was due to the differences in session duration, we analyzed the shared neural dimensions within a 10-minute period during the interaction sessions. Our results showed that shared neural dimensions were present in both GABAergic and glutamatergic neurons during the interaction sessions (Extended Data Fig. 6a,b). Consistent with the longer sessions, the top shared dimensions in GABAergic neurons accounted for a substantially larger fraction of the neural variance compared to glutamatergic neurons (Extended Data Fig. 6c). This indicates that the absence of shared neural dimensions in the separation session was not due to the differences in session duration, and the difference between GABAergic and glutamatergic neurons remained consistent in the shorter time window.

Supplementary Note 5

In both males and females, both GABAergic and glutamatergic neurons contained several shared neural dimensions, and the activity of these shared neural dimensions showed significantly higher-than-chance correlations (Extended Data Fig. 3f-h). In addition, GABAergic neurons contained a larger shared neural subspace across brains compared to glutamatergic neurons in both males and females, while the neural variance captured by the top shared neural dimension was generally higher in males than in females (Extended Data Fig. 3f-h). This sex difference is likely due to the inherent behavioral differences between males and females, with males showing more antagonistic behaviors.

Supplementary Note 6

Although the unique neural dimensions were overall not correlated, they may contain neural population activities that are correlated across animals within specific temporal domains. To examine this possibility, we further investigated the correlation in shared or unique neural population activity in different frequency ranges. By applying the fast Fourier transform to the top shared (PLSC1) and the top unique (U1) neural dimensions (Methods), we found that the correlation of the unique neural population activity was comparable, across timescales, to that of phase-randomized data (Extended Data Fig. 3i).

Supplementary Note 7

Both GABAergic and glutamatergic neurons are broadly distributed along the anterior-posterior and lateral-medial axes. We investigated the anatomical distribution of neurons significantly contributing to the top shared neural dimension by analyzing the average positions of these cells and comparing them to the positions of all recorded cells. Our analysis showed that the average positions of selective cells were largely the same as those of non-selective cells, with minor

deviations along the lateral-medial axis (Extended Data Fig. 6n-q). Specifically, GABAergic neurons contributing to the top shared dimensions were positioned slightly more medially, while glutamatergic neurons were located more laterally, though these differences were relatively small ($<100\ \mu\text{m}$). Given the large size of the dmPFC, it would be intriguing to explore whether the shared and unique neural subspaces vary across its distinct subregions or layers. Furthermore, investigating how these neural subspaces manifest in other brain regions presents an exciting avenue for future research.

Supplementary Note 8

We performed analyses using spike events extracted through the CNMF-E (Extended Data Fig. 4a-f). Consistent with our analysis using fluorescence signals, we found that significantly more shared dimensions emerged during interaction sessions than during separation sessions (Extended Data Fig. 4a, b). Additionally, GABAergic neurons overall showed a greater number of shared neural dimensions than glutamatergic neurons across animals. Approximately 28% of the total neural variance in GABAergic neurons was shared, compared to only 2% in the glutamatergic population (Extended Data Fig. 4c), with the top neural dimension capturing approximately 5 times more neural variance in the GABAergic subpopulation than in the glutamatergic population (Extended Data Fig. 4d). These results are consistent with those observed using fluorescence signals.

Supplementary Note 9

At a slower timescale (10-min), the cosine similarity of the top shared neural dimensions in both cell types was significantly higher than chance between segments (Extended Data Fig. 6d, e). However, at faster timescales (1 min), the shared and unique neural dimensions changed dynamically over time (Extended Data Fig. 6f-i). When examining periods in which animals exhibited similar types of behavioral interactions, we found that the cosine similarity of the top shared neural dimensions in GABAergic neurons was significantly higher than chance (Extended Data Fig. 6j, k). In contrast, the cosine similarity for the unique neural dimensions did not exceed chance levels at either the slow or fast timescales, indicating the highly dynamic nature of the unique neural subspace.

To assess the stability of the top shared neural dimension when the same animal interacted with different partners, we computed the cosine similarity of PLSC1 coefficients for the same animals across different experimental sessions. We found that GABAergic neurons had a similarity score higher than that of the temporally shuffled controls (Extended Data Fig. 7a, b), indicating that the shared neural dimensions are more stable than chance across different animal pairs. Notably, the cosine similarity of GABAergic neurons was greater across animal pairs than that of glutamatergic neurons ($p = 0.0381$, Two-tailed Mann-Whitney test). This suggests that, across different pairs, the shared neural subspace of GABAergic neurons is more aligned than that of glutamatergic neurons.

During mutual social interactions, the same behavior in one animal may elicit different behavioral responses from others. Therefore, we examined how the shared neural subspace is modulated by different behavioral feedback from the interaction partner. We found that the weights of the top shared neural dimensions are overall different, with a few select behavioral pairs (e.g. attack $\leftrightarrow$ escape vs. attack $\leftrightarrow$ defend) showing greater similarity than chance (Extended Data Fig. 7g-i). This suggests that while mutual interactions strongly contribute to inter-brain correlation, different behavioral responses from other animals dynamically shape the shared neural subspace.

Supplementary Note 10

To investigate the relationship between behavioral transitions and the shared neural dimensions, we defined five behavioral states: aggression (attack, chase, tussle, threaten), defense (defend, escape, flinch), general social interaction (general-sniff, sniff-face, sniff-genital, approach, follow), non-social (dig, climb, stand, explore object, bite object) and idle (self-groom, idle). When we fitted these states to the top shared neural dimensions, we found that they explained 14.2% and 45.6% of the total variance of the PLSC1 in glutamatergic and GABAergic neurons, respectively (Extended Data Fig. 6l, m). To determine whether transitions between these states contribute to the top shared neural dimensions, we estimated their unique contributions beyond what was already explained by the behavioral states alone. We built a full model incorporating both behavioral states and their transitions, and compared it to a model using only the behavioral states. The difference in variance explained reflects the unique contribution of behavioral transitions, excluding redundancy with the states. We found that behavioral transitions uniquely contributed 6.4% and 6.5% of the total variance in GABAergic and glutamatergic neurons, respectively (Extended Data Fig. 6l, m). Similarly, for the unique dimension U1, behavioral transitions uniquely contributed 6.6% and 6.2% of the total variance explained in GABAergic and glutamatergic neurons, respectively (Extended Data Fig. 6l, m). These results suggest that while behavioral states have a substantial contribution to the neural subspaces, behavioral transitions also contribute to the shared and unique neural subspace in both cell types.

Supplementary Note 11

To examine potential lag in neural signals between animals, we computed both the covariance and correlation of the top shared neural dimension (PLSC1) across a lag range of -30 to +30 seconds using 100 ms timesteps. We focused on aggressive pairs, comparing dominant animals displaying more attack behavior with subordinate animals displaying more defend and escape behaviors. Our results show that both covariance and correlation of PLSC1 peak at a 0-second lag, with no significant difference within a 1-second window before or after this point (Extended Data Fig. 11g-j). This finding suggests that the shared neural subspace between interacting animals does not exhibit an observable lag under our current experimental setup, likely reflecting rapid behavioral responses between animals.

Supplementary Note 12

While GABAergic cells more strongly represent annotated behaviors than glutamatergic neurons, the encoding of behaviors in glutamatergic neurons is not at chance levels. Therefore, glutamatergic neurons represent some behavioral information, albeit generally less than GABAergic neurons. Extended Data Fig. 10c-d shows the representation of all the raw and derived tracking features in GABAergic or glutamatergic neurons, with GABAergic neurons overall explaining more variance than glutamatergic neurons across all features. These features measure animals' kinematic properties—while they are basic building blocks of social behaviors, they are not equivalent to individual behaviors. We expect that some of these features may be encoded in glutamatergic neurons and contribute to the encoding of behavioral information shown in Fig. 1q-t.

Supplementary Note 13

We trained and evaluated three additional social AI environments featuring varying collaborative and competitive goals and complexities to investigate whether shared neural dimensions also arise in different, more complex AI environments (Extended Data Fig. 13h-t). When agents behaved in a collaborative MARL environment (Extended Data Fig. 13h-j), we found that they displayed significantly more shared dimensions and higher PLSC correlation than the competitive task (Extended Data Fig. 13k, l). These support the notion that collaboration, which often involves a shared goal, can lead to stronger neural coupling between interacting agents. We also implemented a more complex, game theory-based task, known as the Coins task, which can be viewed as a higher-dimensional, spatially extended version of the iterative Prisoner's Dilemma game⁴⁷ (Extended Data Fig. 13m, n). In this task, agents can choose between collaborative and competitive goals. We observed more shared neural dimensions across trained agents than a cross-pair chance level (Extended Data Fig. 13o) (while the number of dimensions cannot be directly compared to the chaser-explorer task due to the difference in the task structure). Lastly, we implemented a significantly more complex social environment (using Melting Pot 2.0) that incorporates more complex visual inputs, dynamic internal states, and a larger action space (Extended Data Fig. 13p, Methods). In this task, successful agents must plan their actions based on complex goals and internal state dynamics, resulting in the emergence of relatively sophisticated strategies and action trajectories (Extended Data Fig. 13q-s). We found that shared neural dimensions emerged in interacting pairs, even though the task and behavior strategies are considerably more complex (Extended Data Fig. 13t). These agents were also trained using an LSTM recurrent neural network, indicating our results are consistent in different recurrent neural network architectures. Together, these results show that despite the increased complexity of these tasks—featuring richer social environments with complex visual inputs, dynamic internal states, and a more diverse action space—inter-agent shared neural dimensions emerged from interaction between artificial agents.

Supplementary Note 14

Beyond modeling biological social behavior, modeling AI social behavior has engineering and societal applications involving real-world interactions with multiple agents, where AI agents engage in distinct tasks and goals to achieve a collective outcome rather than attempting to encompass all capabilities within a single model (e.g., multi-agent robotics). It may also have applications in facilitating the interactions between the AI and humans, such as in shared autonomy “copilot” systems where AIs cooperate with humans to perform tasks or in self-driving, where AIs must interact with both humans and other AI agents.